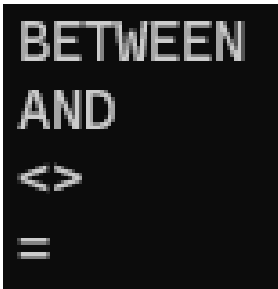


Filtering Data in SQL Lab 2

Summary: In this lab, I practiced filtering numeric and date-based data in SQL to support a security investigation. My role was to act as a security analyst reviewing login activity related to a recent incident. The goal of the lab was to apply comparison operators and filtering techniques to identify suspicious login attempts.

Commands Used:



Overview: To begin the investigation, I started by filtering login activity based on date values. I first retrieved all login attempts that occurred after a certain date by using the `>` operator. This operator returns only the records with a value greater than the one specified, which allowed me to isolate activity that happened following the incident. Using `<`, `>`, and `=` helps compare numeric information, so the query focuses on only the entries that match the criteria.

Next, I narrowed the results to a specific date range by using the `BETWEEN` operator. This command works with two values and returns all records that fall within that range, including boundary values. For example: `SELECT * FROM table_name WHERE date_column BETWEEN 'start' AND 'end';` This made it easier to focus on login attempts that occurred during the exact window of time relevant to the investigation.

I also applied these same filtering techniques to time values and event ID numbers. By using `<` to find logins before a certain time, `BETWEEN` to capture activity within a specific hour, and comparison operators like `>=` to filter event IDs, I was able to narrow the dataset to only the most relevant records for the investigation.

Reflection: Using comparison operators and numerical filters made it much easier to isolate meaningful login activity without manually scanning large tables. These techniques are essential for security analysts because they allow quick identification of unusual patterns, suspicious timestamps, and potentially malicious events. This lab reinforced how SQL operators can efficiently narrow large datasets into focused, actionable insights during an investigation.